When Scale-Fragmenting Regulation Kills Growth

A No-Fold-Crossing Test for Increasing-Returns Sectors

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Abstract

Increasing returns to scale (IRS) are often treated as a warrant for structural intervention. That is too fast. In a companion Ramsey-CES model, IRS imply a minimum viable scale,

$$L_{\min} = \left(\frac{D}{\phi_K^{\max}} \right)^{1/(\theta-1)},$$

below which no interior steady state exists. This paper develops the policy implications and sharpens the claim. First, in a monopolistic-competition decentralization, markups raise the market minimum viable scale to

$$L_{\min}^{dec} = \left(\frac{D}{\alpha \phi_K^{\max}} \right)^{1/(\theta-1)},$$

so conduct remedies that reduce markups can improve feasibility. Second, we formalize a no-fold-crossing test for regulation. A structural remedy is admissible only if the gain from lower markups exceeds the loss of productive complementarities, net of added wedges, by enough to preserve a safety margin from the fold. Third, we reinterpret the evidence accordingly. British rail fragmentation is best understood as a destruction of whole-system complementarities rather than as a generic morality tale about privatization, while the 2025 Google search remedies litigation illustrates the institutional preference for conduct, data-access, and syndication remedies over breakup when scale effects are uncertain. The paper's claim is therefore narrower and stronger than "regulation is bad": in high-IRS sectors, scale-fragmenting regulation is presumptively dangerous unless regulators can show ex ante that it is fold-safe.

Keywords: increasing returns to scale; antitrust; minimum viable scale; structural remedies; scale fragmentation; competition policy

JEL: E13; L13; L40; L51; O41

1. Introduction

Increasing returns to scale characterize many of the sectors that now sit closest to the center of modern growth: software, cloud infrastructure, semiconductors, AI systems, digital platforms, telecommunications networks, rail, and electricity. These sectors share a recognizably similar production logic. They require large sunk investments, exhibit low marginal cost over a wide operating range, and depend on complementarities across assets, data, maintenance, or organizational interfaces.

Policy discussion often compresses these facts into a simple syllogism: increasing returns imply declining average cost; declining average cost implies natural monopoly; natural monopoly implies structural regulation or breakup by default. That syllogism is too coarse for the environments that matter most. It treats IRS as a pricing problem when, in general equilibrium, IRS first create a feasibility problem: whether the economy can sustain the scale needed for high productivity in the first place.

In a companion paper, Milei and Reidel (2026) derive a closed-form minimum viable labor scale,

$$L_{\min} = \left(\frac{D}{\phi_K^{\max}} \right)^{1/(\theta-1)},$$

in a Ramsey-CES economy with IRS. Below that threshold, no interior steady state exists. Above it, two interior steady states may coexist. Small changes in wedges are amplified by the factor $1/(\theta - 1)$, so an intervention that appears modest in partial equilibrium can move the economy sharply toward, or across, the fold.

This paper asks the policy question that follows: **what kinds of regulation are dangerous in an IRS economy?** Our answer is narrower, and stronger, than a blanket anti-regulatory claim. The problem is not regulation as such. It is **scale-fragmenting regulation**: remedies that break productive complementarities without proving that the resulting loss of scale is smaller than the competitive gains they deliver.

That distinction matters because our own framework does not imply laissez-faire. Once the planner's problem is decentralized, markups raise the market minimum viable scale above the planner's level. Conduct remedies that reduce markups can therefore improve feasibility. Properly designed interoperability, data-portability, and open-access rules can do the same. The danger lies in interventions that reduce market power by destroying the productive unit that generates scale in the first place.

The paper makes five contributions.

First, it narrows the empirical claim. We do not need the entire economy to display large increasing returns. We need only the sectors that are most likely to be targets of structural intervention to do so, and a conservative threshold of $\theta \geq 1.25$ is sufficient for our quantitative warnings.

Second, it decentralizes the companion model in a Romer-style monopolistic-competition environment. This yields a transparent formula for the decentralized minimum viable scale and shows that markups move the economy closer to the fold rather than cushioning it from scale constraints.

Third, it formalizes a **no-fold-crossing test**. For any remedy, one can decompose the policy effect into markup relief, complementarity destruction, added wedges, and changes in operating scale, and ask whether the post-remedy economy remains safely away from the fold.

Fourth, it revises the paper's historical evidence. British Rail remains the canonical case, but the right reading is not "all regulation fails" or even "all privatization fails." It is that dense non-contractible interfaces make fragmentation costly and slow to reverse. California electricity is best read in the same spirit: not as proof that deregulation is impossible, but as evidence that badly designed restructuring in a network industry can create amplified feasibility problems.

Fifth, it updates the contemporary application. The 2025 Google search remedies record is informative precisely because it illustrates the distinction between conduct-style remedies and structural separation. Plaintiffs proposed Chrome divestiture and a contingent Android divestiture. The final judgment instead relied on prohibitory injunctions, data-sharing, and syndication remedies while rejecting breakup. Our framework provides a general-equilibrium reason why that difference matters.

The result is a different policy default. IRS do not imply "regulate by default." They imply that regulators must not fragment scale without a factual showing that the remedy is fold-safe.

2. Why sector-specific increasing returns matter more than aggregate averages

For the aggregate U.S. economy, Basu and Fernald (1997) estimate returns to scale only modestly above one. That fact does not resolve the policy question here. Structural antitrust and sectoral regulation do not target the aggregate economy; they target a small subset of firms and industries whose cost structures are unusually heavy in fixed costs, intangible capital, network complementarities, and data feedback.

2.1 Markups and the Romer mapping

In the Romer (1990) variety-expansion model, a competitive final-good sector combines labor with a continuum of intermediates. In symmetric equilibrium, aggregate technology is homogeneous of degree $\theta = 2 - \alpha$, where α is the inverse of the markup, $\mu = 1/\alpha$. This mapping is useful because it links an observable market outcome to the reduced-form scale parameter.

De Loecker, Eeckhout, and Unger (2020) document a broad rise in markups, with technology sectors toward the upper end of the distribution. For the kinds of firms that attract structural scrutiny, markups in the range $\mu \in [1.33, 2.0]$ imply $\alpha \in [0.50, 0.75]$ and therefore $\theta \in [1.25, 1.50]$. That alone is enough to make the fold logic relevant.

2.2 Intangibles, sunk cost, and low marginal cost

The modern intangible economy strengthens the same inference. Haskel and Westlake (2018) document the long rise of intangible investment in business spending. Software, model training, platform engineering, chip design, search indexing, and network software all require large upfront outlays that are not replicated each time output expands. When the relevant operating margin is one more query, one more seat, one more software deployment, or one more packet over an existing network, marginal cost can be very small relative to sunk cost.

A useful heuristic is to think of the production technology as combining a fixed-cost block with a variable-cost block that is closer to constant returns. The larger the fixed-cost share over the relevant operating range, the larger the effective returns to scale. We do not need to know that the true θ is huge. We need only to know that, for the sectors most likely to be broken up, a calibration such as $\theta = 1.25$ is not aggressive.

2.3 Network industries and vertical complementarities

Network industries add a further margin that simple markup stories miss: productive complementarities across tightly linked tasks. A railway does not consist of separable lines of business that can be recombined frictionlessly by contract. Track, signaling, maintenance, timetabling, rolling stock, and safety management interact continuously. Electricity systems combine generation, dispatch, transmission constraints, reliability standards, and retail interface rules. Telecommunications and digital platforms exhibit related cross-unit dependencies.

These complementarities matter because the reduced-form object $\phi_K^{\max}$ is not just a measure of physical capital productivity. It captures the peak marginal productivity of capital **given the organization of production**. Destroying interfaces, dispersing knowledge, or making coordination contractual rather than internal can lower $\phi_K^{\max}$ even when all the physical assets remain in place.

Antweiler and Trefler (2002) estimate returns to scale around 1.1-1.2 for manufacturing. The sectors under structural regulatory pressure are often more extreme than that, not less. Their production logic is dominated by sunk design, software, compute, search, data, or network interface costs rather than by easily replicable shop-floor inputs.

2.4 A conservative working threshold

The key point is methodological. The relevant scale parameter for policy is not the industry average; it is the scale parameter of the specific productive unit that the proposed remedy would fragment. For that reason, the paper adopts $\theta = 1.25$ as a conservative benchmark rather than as an aggressive central estimate.

That benchmark is demanding enough to matter but modest enough to be defensible. It also makes the policy test stricter in the right direction. Because the amplification factor is $1/(\theta - 1)$, even modest IRS can generate large policy effects. If θ is larger than 1.25 in the relevant sector, the same qualitative warning remains. If θ is only slightly above one, the amplifier becomes even larger.

3. Model summary and the companion results

We summarize only the results needed for the policy argument; proofs and calibration details are in Milei and Reidel (2026).

A planner chooses consumption and labor to maximize

$$\sum_t \beta^t \left[\eta \log C_t + (1-\eta)\log(1-L_t)\right]$$

subject to

$$C_t + K_{t+1} = (1-\delta)K_t + F(K_t, L_t),$$

with CES technology

$$F(K, L) = A\big(\omega K^{\xi} + (1-\omega)L^{\xi}\big)^{\theta/\xi}, \qquad \theta > 1.$$

Let the user-cost wedge be $D = \beta^{-1} - (1-\delta)$. In the fold regime, the reduced marginal product of capital is hump-shaped, with peak $\phi_K^{\max}$. The steady-state Euler condition implies

$$L \geq L_{\min} = \left(\frac{D}{\phi_K^{\max}}\right)^{1/(\theta-1)}.$$

Below $L_{\min}$ no interior steady state exists. Above it, two interior steady states typically coexist. This already makes clear why partial equilibrium can mislead: policies that shift D or $\phi_K^{\max}$ do not simply move prices; they can change whether the high-scale equilibrium exists at all.

A useful comparative-static identity is

$$\Delta \ln L_{\min} = \frac{\Delta \ln D - \Delta \ln \phi_K^{\max}}{\theta - 1}.$$

The denominator creates an amplifier. Small changes in wedges become large changes in minimum viable scale.

Table 1 reports the illustrative calibration from the companion paper.

Table 1: Illustrative amplification of the minimum viable scale

Shock to D	New $L_{\min}$	Change in $L_{\min}$
+10%	0.37	+46%
+25%	0.61	+144%
+40%	0.96	+284%

In the baseline calibration ($\theta = 1.25, \sigma = 2/3, A = 0.057$), the high- and low-scale steady states differ sharply. Output differs by roughly a factor of seven, and the consumption-equivalent welfare gap exceeds 680 percent. These numbers are calibration-dependent, but the policy lesson is not: in the fold region, the economy can be fragile to seemingly small structural or regulatory changes.

4. Decentralization and a no-fold-crossing test for regulation

4.1 Why competitive equilibrium fails under IRS

With $\theta > 1$, a price-taking firm facing constant factor prices has convex profits in scale. Competitive equilibrium therefore fails in the standard way: the firm wants to expand without bound. The familiar resolution is monopolistic competition. Each producer faces a downward-sloping demand

curve over a differentiated variety and earns a markup over marginal cost.

That institutional detail matters because it changes the policy baseline. If markups are part of the mechanism that allows IRS production to be organized at finite scale, remedies must be evaluated relative to that decentralized equilibrium rather than relative to the planner.

4.2 The decentralized minimum viable scale

Use a Romer-style decentralization. A competitive final-good sector combines labor L with a continuum of intermediates $x(i)$:

$$Y = L^{1-\alpha} \int_0^N x(i)^\alpha di, \qquad \alpha \in (0,1).$$

Each intermediate is produced by a monopolist using capital, and the optimal markup is $\mu = 1/\alpha > 1$. In symmetric equilibrium with variety creation proportional to capital, the aggregate technology is homogeneous of degree $\theta = 2 - \alpha$.

The household Euler equation pins down the private rental rate at $r = D$, but the social marginal product of capital is higher by the markup wedge. Equivalently, the decentralized economy behaves as if it faced a higher effective hurdle rate. The resulting minimum viable scale is

$$L_{\min}^{dec} = \left(\frac{D}{\alpha \phi_K^{\max}} \right)^{1/(\theta-1)} > L_{\min}^{planner}.$$

Proposition 1 (Markups raise minimum viable scale). Since $\alpha < 1$, the decentralized economy is closer to the fold than the planner's allocation.

The intuition is the opposite of a common partial-equilibrium reflex. Market power does not necessarily "finance" fixed costs in a way that makes the system safer. In this reduced-form setting, the markup lowers the private return to capital relative to its social marginal product and thereby raises the threshold scale needed to sustain the high-productivity equilibrium.

4.3 Comparative statics for remedies

For policy analysis, the most useful log-linear decomposition is

$$\Delta \ln L_{\min}^{dec} = \frac{\Delta \ln D - \Delta \ln \alpha - \Delta \ln \phi_K^{\max}}{\theta - 1}.$$

This says that a remedy can affect feasibility through three channels:

1. **The wedge channel** (D): compliance costs, financing frictions, regulatory uncertainty, or operational rigidity can raise the effective user cost of capital.
2. **The markup channel** (α): remedies that reduce market power increase α and lower minimum viable scale.
3. **The complementarity channel** ($\phi_K^{\max}$): remedies that fragment productive organization reduce the peak marginal productivity of capital and raise minimum viable scale.

The complementarity channel is the crucial one for structural remedies. A breakup can lower markups and still be harmful if it lowers $\phi_K^{\max}$ even more.

4.4 The no-fold-crossing test

Let L_{obs} denote observed **effective operating scale**. In a standard industry application, this can be literal labor. In digital or network settings, it may be a labor-equivalent scale measure that embeds traffic, query volume, user base, or compute capacity once those inputs are mapped into the reduced-form technology.

Define pre-remedy headroom from the fold by

$$h_0 = \ln\left(\frac{L_{obs}}{L_{\min,0}^{dec}}\right).$$

Now consider a remedy r. Let

- $m_r = \Delta \ln \alpha_r$ be markup relief,
- $c_r = -\Delta \ln \phi_{K,r}^{\max} \geq 0$ be complementarity destruction,
- $u_r = \Delta \ln D_r \geq 0$ be added wedge pressure,
- $\ell_r = \Delta \ln L_{obs,r}$ be the change in effective operating scale,
- $s > 0$ be the regulator's safety buffer.

Then the remedy changes minimum viable scale by

$$\Delta \ln L_{\min}^{dec}(r) = \frac{u_r - m_r + c_r}{\theta - 1},$$

and post-remedy headroom is

$$h_r = h_0 + \ell_r - \frac{u_r - m_r + c_r}{\theta - 1}.$$

No-fold-crossing test. A remedy is admissible only if

$$h_r \geq s.$$

Equivalently,

$$c_r \leq m_r - u_r + (\theta - 1)(h_0 + \ell_r - s).$$

This is the paper's core policy rule. The right question is not whether a remedy reduces market power in some local partial-equilibrium sense. The right question is whether, after accounting for the destruction of productive complementarities and any additional wedges, the economy remains safely away from the fold.

4.5 A robust version under uncertainty

Real policy is never estimated without error. Let conservative bounds be $m_r \geq \underline{m}$, $c_r \leq \overline{c}$, $u_r \leq \overline{u}$, $\ell_r \geq \underline{\ell}$, and $\theta \geq \underline{\theta} > 1$. A sufficient condition for safety is then

$$h_0 + \underline{\ell} - \frac{\overline{u} - \underline{m} + \overline{c}}{\underline{\theta} - 1} \geq s.$$

This robust version turns the policy rule into a burden-of-proof principle. A regulator proposing fragmentation should have to show that the remedy clears the safety threshold under conservative assumptions, not merely under a favorable point estimate.

4.6 A three-way taxonomy of remedies

The decomposition above yields a natural taxonomy.

Table 2: Remedy taxonomy in IRS sectors

Remedy type	Effect on α	Effect on $\phi_K^{\max}$	Typical effect on $L_{\min}^{dec}$	Presumption
Conduct remedies (exclusive dealing, tying, self-preferencing, discriminatory contracts)	$\uparrow$	approximately unchanged	Down	Favorable
Scale-preserving access remedies (interoperability, data portability, open access)	$\uparrow$	small or zero if well designed	Usually down	Favorable with implementation check

Scale-fragmenting structural remedies (breakups, forced separation, network-size caps, bans on integration)	↑	↓	Ambiguous; often up	Disfavored unless fold-safe

The legal label is not what matters. A conduct remedy can behave like a structural fragmentation if the prohibited conduct is itself part of the productive complementarity. Segal and Whinston (2007) make exactly this point in the context of innovative industries. The correct discipline is therefore functional: map the remedy into m_r, c_r, u_r, and ℓ_r, then apply the test.

5. Historical and contemporary applications

5.1 British Rail: fragmentation, interfaces, and hysteresis

British Rail remains the clearest institutional example of the mechanism we have in mind. The early-1990s reform split British Rail into **over 100 private companies** (House of Commons Library, 2017). That alone does not prove failure. What matters is the density of interfaces that the fragmentation created.

The McNulty review, commissioned by the UK government, is especially useful because it is not ideological. It finds that among the principal barriers to efficiency in Great Britain's rail system are the **fragmentation of structures and interfaces**, misaligned incentives, and a resulting "co-operation deficit" that makes it hard to secure whole-system solutions (McNulty, 2011). The same review reports that passengers in Great Britain were paying around 30 percent more than comparable European benchmarks and that the UK taxpayer was likely paying at least 30 percent more as well.

This maps naturally into the model. Fragmentation can raise α by introducing more contractual contestability and reducing integrated control. But if it also destroys the complementarities among infrastructure management, maintenance, timetable planning, and safety stewardship, it lowers $\phi_K^{\max}$. McNulty's language is precisely the language of a fall in $\phi_K^{\max}$: too many interfaces, misaligned incentives, and barriers to "whole-system" optimization.

The safety record should also be stated carefully. It is too crude to say simply that "safety collapsed." The stronger and more accurate statement is that the fragmented regime revealed serious coordination weaknesses. The Office of Rail and Road's 2020 strategy paper states that the Hatfield derailment "illustrated the inadequacies of Railtrack and their contractors' arrangements" and that the aftermath revealed unacceptable deterioration in track condition. Network Rail later brought maintenance back in-house and undertook major improvements (Office of Rail and Road, 2020). That sequence is exactly what one would expect if the fragmented system had overshot into too much interface dependence and then tried, slowly and imperfectly, to rebuild the lost complementarity.

The ownership history reinforces the same interpretation. An official Department for Transport evidence paper notes that Railtrack was placed into railway administration on 7 October 2001, with Network Rail eventually taking over its responsibilities (Department for Transport, 2019). The UK's 2025 rail reform response still states that "structural fragmentation" inhibits how the railway is run and justifies a new coordinating body, Great British Railways (Department for Transport, 2025).

None of this proves that every vertically integrated railway must outperform every separated one. But it is unusually strong evidence for the narrower claim that, in a high-interface network industry, fragmentation can impose large and durable losses in productive complementarity. That is enough for our purpose.

5.2 California electricity: a cautionary case of bad restructuring

The California electricity crisis is informative for a related reason. Joskow (2001) explicitly warns against reading the crisis as proof that all electricity restructuring was doomed. The stronger lesson is that California adopted a radical restructuring design and then combined it with pricing and procurement rules that made the system brittle. By 2001, the state had effectively taken over key parts of the power sector and replaced the nascent competitive market with state procurement and regulated retail prices.

That is precisely the type of case our framework is designed to interpret. In a network industry with reliability constraints and nontrivial operational coordination, separation can reduce productive complementarities while poorly designed pricing rules raise effective wedges. The crisis therefore fits the model as a case of $c_r > 0$ and $u_r > 0$ rather than as a generic referendum on competition policy.

5.3 Google and the choice of remedy in a digital platform case

The Google search remedies record is the best contemporary application because it forces the policy choice into the open. In March 2025, the United States and plaintiff states summarized their revised proposed final judgment by emphasizing three elements: an end to search-related payments, a requirement that Google divest Chrome, and a contingent Android divestiture. The same filing described Chrome as "a critical search access point through which more than 30 percent of search inquiries are routed" (U.S. Department of Justice, 2025a).

By December 2025, however, the court's Memorandum Opinion and Final Judgment had taken a different shape. The court adopted a package centered on prohibitory injunctions, compelled sharing of certain data, syndication remedies, and a Technical Committee. It rejected more severe proposals, including Chrome divestiture, mandated choice screens, and a complete payment ban (U.S. District Court for the District of Columbia, 2025).

That record is important even if one takes no view on Google's underlying liability. It shows that real-world remedial choice in digital platform markets is not a binary between "do nothing" and "break the firm up." There exists a large middle category of remedies aimed at exclusionary contracts, access to rivals, and market opening without immediate structural dismemberment.

Our framework gives a general-equilibrium interpretation of that difference. Payment restrictions and syndication or data-sharing remedies are Category 1 or Category 2 instruments if they reduce exclusion while leaving the productive unit largely intact. Chrome or Android divestiture is Category

3 if it materially reduces productive complementarities among distribution, query flow, indexing, model improvement, and adjacent services. The point is not that breakup is never justified. It is that, in an IRS sector, breakup should not be presumptive. It should require a factual showing that c_r is small enough to pass the no-fold-crossing test.

The court's language points in the same direction. In explaining why it rejected some requested remedies, the court emphasized that "there can be no remedy absent a factual basis to support it" (U.S. District Court for the District of Columbia, 2025). Our contribution is to give that general requirement a specific economic content for IRS sectors: the factual basis must include the remedy's likely effect on minimum viable scale.

6. Policy implications

6.1 IRS are not a monopoly theorem

Increasing returns to scale are a technological property; monopoly is a market-structure outcome. The two are related but not identical. A sector may exhibit strong increasing returns and still sustain rivalry through differentiated products, contestability, open standards, or multiple firms operating at efficient scale. Conversely, a sector can exhibit market power for reasons that have little to do with IRS, including regulation, patents, switching costs, or strategic exclusion.

This distinction matters because the first-order implication of IRS in our framework is not an abstract deadweight-loss triangle. It is the existence of a minimum viable scale and the possibility of fold dynamics. That perspective is closer in spirit to Demsetz's (1973) insistence on separating technology from market structure and to Williamson's (1968) warning that anti-monopoly policy can sacrifice productive efficiency if it treats scale as inherently suspect.

6.2 A scale impact assessment

The no-fold-crossing test yields a practical institutional proposal: any structural remedy in a sector with plausible IRS should be accompanied by a **Scale Impact Assessment (SIA)**. At a minimum, the SIA should do five things.

1. Estimate a conservative range for θ for the relevant productive unit, not just for the broad industry.
2. Identify the complementarities that support $\phi_K^{\max}$: engineering coordination, data flows, maintenance interfaces, logistics, standards, compute integration, or network management.
3. Estimate the remedy's effect on markups, m_r.
4. Estimate the remedy's effect on productive complementarity, c_r, on wedge pressure, u_r, and on effective operating scale, ℓ_r.
5. Apply the robust no-fold-crossing test using conservative bounds and an explicit safety margin s.

This does not forbid intervention. It disciplines it. In that sense, the proposal is analogous to an engineering safety check or an environmental impact assessment: not a veto, but a requirement that irreversible decisions be justified before they are taken.

6.3 Burden of proof for structural fragmentation

Once the safety margin is made explicit, the burden of proof becomes clear. Conduct remedies should generally be judged against the usual evidentiary and proportionality standards. Scale-fragmenting remedies should face a stricter burden because their downside is amplified and potentially irreversible.

Put differently: if a regulator cannot show that the remedy passes the robust no-fold-crossing test under conservative assumptions, the regulator has not yet made the case for structural separation. The stronger the underlying IRS, the stronger this presumption should be.

This is also the most defensible way to make a strong argument against aggressive regulation. The strongest case is not "all regulation is destructive." That claim is both empirically and theoretically false. The strongest case is that **regulators are not entitled to gamble with productive scale in IRS sectors when the gamble is asymmetric and hard to reverse.**

6.4 Target conduct, preserve productive scale

The policy priority that follows is straightforward.

First, enforce vigorously against exclusionary conduct. Exclusive distribution, tying, discriminatory access terms, self-preferencing, and foreclosure contracts can entrench market power while doing little or nothing to increase productive complementarities. Remedies that attack those practices directly often lower $L_{\min}^{dec}$.

Second, prefer scale-preserving opening measures where possible. The Digital Markets Act's interoperability, data-portability, and business-user data-access obligations are examples of remedies that are at least conceptually closer to Category 2 than to Category 3. Their rationale is to reduce barriers and markups without presuming that the productive unit must be dismantled.

Third, reserve structural separation for cases where the regulator can show that complementarities are either weak, replicable, or themselves inseparable from the exclusionary conduct being condemned. Even then, the burden should remain on the regulator to show fold safety, not on the defendant to disprove it.

6.5 Hysteresis and the asymmetry of error

A final implication is temporal. When the economy crosses the fold, error is not symmetric. Under the companion model, recovery requires more than a legal reversal. The wedge must be reduced sufficiently and the economy must be placed back inside the local activation region. In real institutions, that means rebuilding teams, supplier relationships, operating routines, data pipelines, engineering practices, and the credibility of the coordinating center.

British rail is the right cautionary example here. Even after years of reform and reintegration, the UK government still describes fragmentation as a live problem. That is what hysteresis looks like outside the model.

7. Conclusion

The most defensible anti-regulatory claim in an IRS economy is not that regulation always harms growth. It is that **scale-fragmenting regulation is presumptively dangerous unless regula-**

tors can demonstrate ex ante that it is fold-safe.

That claim follows directly from the companion model once the planner's economy is decentralized. Markups raise minimum viable scale above the planner's threshold, so some forms of regulation are clearly beneficial. But remedies differ. Conduct remedies and well-designed access rules can lower the decentralized threshold. Structural fragmentation can lower it, leave it unchanged, or raise it. The outcome depends on whether the loss of productive complementarities is small enough relative to markup relief and available headroom.

The paper's policy rule is therefore simple: do not ask only whether a remedy lowers market power. Ask whether it preserves enough distance from the fold. In high-IRS sectors, that is the feasibility question that should come first.

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